

PULSE: A Screenshot-Based Labeling Tool for Investigating Smartphone Behavior, Intent, and Perception

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Abstract

Understanding mobile phone behavior has attracted significant attention in HCI and mobile health research. While experience sampling methods (ESM) are widely used, they primarily offer discrete snapshots, making it difficult to observe behavioral flow, transitions, and differences between phone use associated with varying intents and levels of habitualness. We introduce **Pulse** (Phone Use Labeling via ScrEenshot), a research application that integrates screenshot logging, passive sensor data collection, and a screenshot-based labeling interface designed to support efficient labeling workflows. The app also supports micro-ESM delivery and a dedicated *researcher mode* allows for flexible configuration and testing of study parameters. We report a case study deployment of *Pulse*, through which we collected approximately 1.65 million labeled screenshots from 25 participants. Preliminary analysis reveals associations between users' phone use intents and their subsequent evaluations of time use, demonstrating the utility of screenshot-based labeling for understanding mobile behavior in context.

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UbiComp Companion '25, Espoo, Finland

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ACM ISBN 979-8-4007-1477-1/2025/10
<https://doi.org/10.1145/3714394.3754395>

CCS Concepts

• **Human-centered computing** → Ubiquitous and mobile computing systems and tools.

Keywords

Smartphones, mobile, screenshot, labeling, intent, time use

ACM Reference Format:

Je-Wei Hsu, Ching-Ting Lin, Uei-Dar Chen, Jui-Ching Kuo, Jui-Chun Liu, Yong-Han Lin, Chen-Ya Chen, Razieh Pourafshari, Yifei Lu, Joseph Bayer, and Yung-Ju Chang. 2025. PULSE: A Screenshot-Based Labeling Tool for Investigating Smartphone Behavior, Intent, and Perception. In *Companion of the 2025 ACM International Joint Conference on Pervasive and Ubiquitous Computing (UbiComp Companion '25)*, October 12–16, 2025, Espoo, Finland. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3714394.3754395>

1 Introduction

As smartphones have become deeply embedded in everyday routines, researchers have increasingly sought to understand how mobile technologies are used in situ. Studying mobile phone behavior offers critical insights, for example, into behaviors around app use [9, 10, 15, 18], mobile media consumption [4, 13, 16], notification interaction [5, 6, 12, 14, 19], etc. To characterize such behaviors, researchers often rely on the Experience Sampling Method (ESM) [8, 20], which captures users' activities or situations at sampled moments throughout the day.

While ESM is effective for capturing self-reported experiences at specific moments [8, 20], the data it produces are inherently

discrete and temporally fragmented. This discontinuity constrains our ability to observe the flow of mobile phone use over time or to understand how users transition between different states of phone use. For example, ESM may capture that a user was reading news at one moment and checking messages at another, but it does not reveal what occurred between those moments or how one activity led to the next. As a result, it is challenging to reconstruct sequential behavioral patterns from ESM data, or to distinguish between recurrent, habitual phone use and goal-directed, incident-driven interactions.

To enable researchers to move beyond isolated moments and capture extended patterns of mobile phone use, we developed *Pulse*, a mobile research tool that allows participants to label periods of phone activity through a screenshot-based interface. *Pulse* automatically captures screenshots during active phone use and enables participants to label each usage session in terms of their intent, perception of time use, and whether the behavior was habitual or goal-directed. To support labeling workflow, the app provides a dedicated interface equipped with features designed to improve labeling efficiency and reduce participant burden.

In addition to screenshot capture, the app logs passive sensor data and enables micro-ESM delivery. Furthermore, a *researcher mode* is included to allow flexible testing and reconfiguration of study parameters.

This paper presents the core features of *Pulse*. We also report a case study deployment in which we used *Pulse* to collect real-world smartphone usage data from 25 participants, who collectively contributed 1,649,936 labeled screenshots. We present results from a preliminary analysis of this dataset. For instance, we found that phone use associated with an entertainment or no specific intent was most frequently labeled as bad use of time, whereas phone use associated with information-seeking intent were most frequently labeled as good use of time. These examples highlight the value of linking users' intent with their evaluation of time use on their phone, offering insights into how perceptions of time well spent vary by usage intent.

2 Pulse: A Research Tool for Labeling and Investigating Smartphone Behavior, Intent, and Perception

Pulse is an Android-based research application designed to allow participants to label their phone use through a screenshot-based interface. Below, we describe the core components and functionalities that support its data collection and labeling capabilities.

2.1 Data Collection by Pulse

Pulse collects three key types of data: phone screenshots, sensor data, and micro-ESM reports. By default, screenshots and sensor data are captured at one-second intervals between 8:00 a.m. and 11:00 p.m., as long as it spans at least 12 hours. Participants can adjust their preferred hours for data collection. The app does not capture screenshots when the screen is off or when users are within *Pulse* itself. In parallel, the app logs phone sensor data—such as app usage, transportation mode, and network state—along with interaction logs. Interaction logs capture events indicating user engagement, including screen on/off status, notification interactions, and events

recorded through Android's accessibility services (e.g., click, scroll, hover, select, or text input).

2.2 Screenshot Labeling

Pulse includes a screenshot labeling interface that organizes screenshots into five daily *periods*, automatically segmented based on the user's customized data collection window (see Figure 1 a). This design is to allow participants to label their phone use sessions throughout the day, rather than in a single batch at the end of the day, thereby reducing burden and minimizing recall errors. *Pulse* sends labeling prompts when a period has been prepared for labeling. We chose the screenshot labeling approach not only to capture more fine-grained temporal data, but also because screenshots help participants recall past activities and reconstruct working contexts [11].

Within each period, screenshots are displayed in chronological order and presented in a grid layout. Participants can adjust the number of screenshots shown per row (up to 10), allowing them to balance visual detail and overview. Screenshots are grouped into *sessions*. A session represents continuous segments of phone use. A session ends when either of the following conditions is met: (1) the screen is turned off for more than 30 seconds, (2) no interaction is detected within 30 seconds, or (3) the user enters *Pulse*. The definition of a session and the 30-second threshold are adopted from prior work [3, 17]. Session breaks are visually indicated (as a clock icon) in the interface along with their durations, helping participants recognize boundaries between distinct phone use sessions.

Researchers can choose to display all captured sessions or to enable session sampling. To ensure diversity in labeled behaviors, *Pulse* adopts a stratified sampling approach that bins sessions into six duration categories: 0–5s, 5–30s, 30–60s, 60–180s, 180–300s, and 300+s. Each bin has a defined sampling quota. This is to ensure a balanced representation of short, medium, and long phone use sessions. These bin definitions are informed by prior research on phone usage patterns and inter-session gaps [1, 2, 21]. Researchers can configure the maximum number of sessions sampled from each bin. In our case study, we set a maximum of two sessions per bin per period, allowing up to 12 labeled sessions per period and up to 60 labeled sessions per participant per day.

Researchers can define the categories to be labeled according to their study goals. In our case study, participants labeled each session along four dimensions: *usage intent*, *perceived urgency*, *time evaluation*, and *goal/habit orientation* (see Figure 1b). To label screenshots, participants select a range by tapping the first and last screenshots within a session—rather than using drag-and-scroll interactions, as employed in prior tools [7]. Labels are then applied by tapping the corresponding label button. Once labeled, the selected range is visually marked with the applied labels (Figure 1c). This start-to-end range selection interaction improves labeling efficiency, particularly for extended sessions such as video watching. Participants can flexibly switch between label categories and apply them in any order.

To further increase the efficiency of the labeling process, the app reduces visual redundancy by stacking consecutive screenshots that are nearly identical, based on computed perceptual similarity [22]. This prevents users from having to scroll through numerous

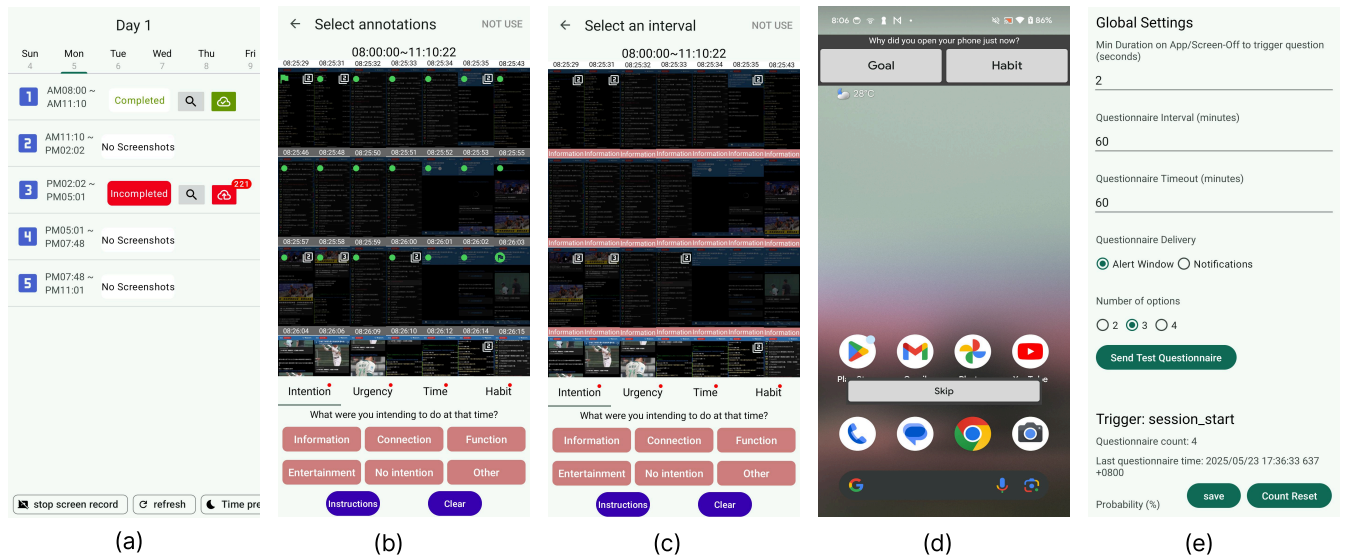


Figure 1: User interfaces for the main functions of Pulse Interface

repetitive screenshots, thereby minimizing effort during labeling of prolonged, uniform activities. In addition, observations from our pilot study revealed that participants sometimes overlooked session breaks when scrolling quickly. To address this, Pulse automatically pauses scrolling when the viewport approaches a session boundary, helping participants notice potential shifts in usage and apply more accurate labels.

2.3 Additional Features

2.3.1 Privacy-Preserving Mechanisms. All collected screenshot data remain on the participant’s device until explicitly approved for upload. The upload interface mirrors the labeling interface, allowing participants to review their screenshots and selectively exclude any they prefer not to share with the research team. This design ensures participants retain full control over their data and may withhold screenshots containing sensitive or personal content before submission to the secure data server. Additionally, both screenshot capture and data upload functionalities can be paused or disabled by the user at any time.

2.3.2 Micro ESM. Pulse also supports the delivery of micro experience sampling (Micro ESM), as illustrated in Figure 1d. Micro ESM prompts appear as dismissible pop-ups and are designed to minimize user burden. The system delivers up to 15 prompts per day, spaced at least one hour apart. Each prompt includes lightweight response options that require no more effort than dismissing the message, thereby reducing the likelihood of habitual dismissal. Micro ESM is currently configured to be triggered at key transition points in phone use: (1) at the start of a phone use session, (2) immediately after a session concludes, and (3) when the user switches between apps.

2.3.3 Researcher Settings. Pulse includes a dedicated researcher mode that provides access to additional settings for study deployment and experimentation. Researchers can adjust parameters such

as period and session segmentation, screenshot similarity thresholds, labeling categories and options, and micro-ESM settings (see Figure 1e for an example). This flexibility supports iterative testing during pilot studies, as well as adaptation of the app to suit diverse research designs and objectives.

3 Case Study: How Users’ Perception of Time Use Varies by Phone Use Intent

We have used Pulse to collect a labeled screenshot dataset from 25 smartphone users and conducted a preliminary analysis to demonstrate how the tool can be used to investigate mobile phone usage patterns. In the following sections, we describe the data collection process and present example analyses that illustrate how users’ perceptions of time use differ depending on their phone use intent.

3.1 Recruitment and Participants

We recruited participants through a market research company to ensure a demographically diverse and balanced sample. The recruitment criteria were as follows. Participants had to be Android smartphone users aged 18 or older. Due to compatibility issues, eligible devices needed to run Android versions 10 to 14. Certain smartphone brands known to be incompatible with the app were excluded. Since screenshots were temporarily stored on the device, at least 10 GB of available internal storage was required. Participants also needed daily access to Wi-Fi or mobile data so researchers could remotely monitor data collection progress. A total of 32 participants were initially enrolled. Of these, one withdrew before data collection began, two were unable to complete the study due to phone-related issues, and four were excluded due to low compliance with study procedures. The final sample consisted of 25 participants (11 male, 14 female), ranging in age from 24 to 57 years ($M = 36.0$, $SD = 10.52$), and representing a variety of occupational backgrounds. Among these 25 participants, the average time spent on labeling per period was 167 seconds ($M = 111.0$, $SD = 189.9$).

3.2 Study Procedure

All participants attended a pre-study orientation session in person. During the session, researchers explained the study’s background and objectives. They introduced the labeling categories, providing definitions and examples for each. Participants were also informed of their responsibilities during the study period. Because screenshots may contain confidential or sensitive information, researchers explained why the data were being collected, how they would be used, and how they would be kept secure and confidential. Participants were also shown how to pause screenshot recording, opt out of data sharing, and permanently delete any collected data from their device.

After obtaining informed consent, we assisted participants in installing the study application on their phones. A test labeling prompt was sent to allow researchers to guide participants through the labeling process and ensure they were familiar with the interface. Participants were informed that they were expected to label all visible screenshots captured by the app each day. Upon receiving a labeling prompt, participants had up to six hours to complete the labeling task.

Participants were required to complete 14 days of data collection. Researchers monitored the uploaded data to ensure proper app functionality and proactively contacted participants if any data issues were detected. Participants who successfully completed the full data collection received a compensation of NT\$2000 (approximately US\$65). The study was approved by our university’s Institutional Review Board (IRB).

3.3 Data Cleaning and Analysis

Among the 25 participants, a total of 23,799 phone usage sessions were detected within our app, of which 51.5% (12,161) were sampled. After data cleaning and noise removing, 21,645 phone sessions were included in the final analysis. A total of 3,129,445 screenshots were collected. Of these, 56.3% (1,762,028) were sampled for labeling. We excluded screenshots with missing labels in any of the four categories, as well as those where the user was not actively using their phone or labeled their intention as "other." After filtering, 93.6% (1,649,936) of the sampled screenshots remained for analysis.

4 Results

In terms of the number of screenshots labeled under each intent category—entertainment accounted for the largest share, comprising 43.5% of all labeled screenshots. This was more than twice the duration of any other intent. Functional use constituted 20.4% of the labeled screenshots, followed by social connection (15.4%). Information-seeking activities made up 11.6%, while phone use with no specific intent accounted for 9.2%. Sessions labeled with unspecified intent tended to be brief, which may explain their lower cumulative duration in the dataset.

Phone use labeled as *entertainment* or *no specific intent* was least likely to be perceived as good use of time—only 26.0% and 26.4% of the time, respectively. In contrast, *functional use* (44.4%) and *social connection* (43.6%) were perceived as good use of time in nearly two-fifths of the cases. The intent most frequently perceived as good use of time was *information seeking*, reported as such in 61.9% of the cases. Notably, for these three latter intents—functional use,

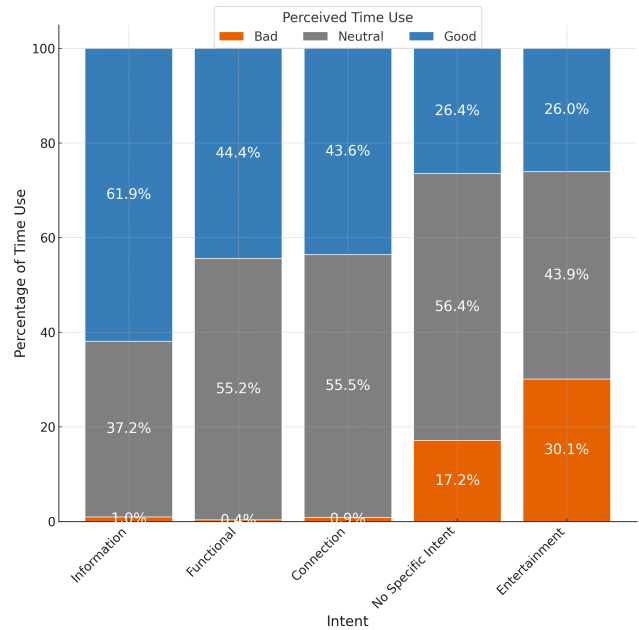


Figure 2: The relationship between Intent and Timeuse

social connection, and information seeking—participants rarely perceived them as bad use of time. In contrast, *entertainment* was perceived as bad use of time in 30.1% of the cases, the highest rate among all intent categories. This result contrasts with findings from [9], which reported that entertainment-related phone use was less frequently associated with regret compared to phone use with no specific intent. We speculate that while users may not feel regretful about engaging in entertainment, they may still believe that such time could have been better spent. This distinction suggests that entertainment use—though emotionally satisfying—may still present opportunities for intervention aimed at promoting more valued or goal-aligned phone use.

5 Future Work

Building on these findings, our future work includes a deeper investigation of how users evaluate their phone use across the spectrum of habitual versus goal-directed activities. In addition, we plan to apply multimodal large language models (MLLMs), as employed in [9], to analyze the sequential patterns of screen activity associated with different intents, levels of habitualness, and perceived time use. This will allow us to explore how visual and contextual cues present in screenshot sequences may inform users’ retrospective evaluations, and to uncover more nuanced behavioral trajectories that underlie digital habits and task-oriented interactions.

Acknowledgments

We gratefully acknowledge the support provided by the National Science and Technology Council, Taiwan (113-2423-H-A49-003, 113-2222-E-A49-009-MY3). We also extend our sincere thanks to all the study participants, reviewers, and colleagues, whose support and feedback were instrumental in the completion of this research.

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